

Dr. Umar Rashid

Post-doctoral Researcher | Molecular Electronics & Instrumentation

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Physikalisches Institut (PHI), Karlsruhe Institute of Technology (KIT), Germany

Research Profile

Experimental physical chemist and instrument designer with expertise in molecular electronics, spectroscopy, and nanoscience. Skilled in developing instrumentation and programming custom tools for charge transport studies at the nanoscale. Combines a strong background in chemistry, physics, biology and engineering to explore fundamental and applied aspects of single-molecule devices. Designed and built large area EGaIn based charge transport setup, Mechanically controlled break junction(MCBJ) and Scanning tunneling microscope break junction(STM-BJ) setup for single molecular measurements. Published in *JACS* and *Angewandte Chemie*, with multiple cover features and national/international awards.

Education

- **Ph.D. in Chemistry (2025)** — Indian Institute of Science, Bangalore
Prime Minister's Research Fellow | Thesis on Charge Transport in Molecular Junctions
 - **Research Training Program (2019)** — Indian Institute of Science, Bangalore
CGPA 9.3/10 | Best Performance Award
 - **M.Sc. in Chemistry (2018)** — University of Kashmir
Gold Medalist | Rank 1 in Entrance Examination
 - **B.Sc. (2015)** — University of Kashmir
1st Rank, Chemistry, Zoology, Botany, English
 - **Secondary Education (2012)** — J&K State Board of School Education
Science stream - Physics, Chemistry, Biology, Environmental Science
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Current Position

Post-doctoral Researcher, Physikalisches Institut (PHI), Karlsruhe Institute of Technology (KIT), Germany - (From August, 2025)

Research Interests

- Instrumentation and methodology development for molecular electronics
- Stimuli-controlled transport (light, bias, temperature, magnetic field)
- Charge transport through single molecules and self-assembled monolayers
- Integration of chemistry, physics, and materials science for nanoscale device design

Technical Expertise

Experimental & Analytical:

- UV-Vis, NMR, FTIR, Fluorescence spectroscopy
- Electrochemical methods (CV, EIS, potentiometry)
- X-ray Diffraction, XPS, UPS
- AFM/STM, MCBJ, EGaIn Junctions
- Fabrication techniques - PVD, ALD, Sputtering, Chemical and Plasma etch

Programming & Design:

- LabVIEW (Professional), Python (Basic), MATLAB, OriginLab
- 3D Design: AutoCAD Inventor, SolidWorks, 3D Printing

- Blender, Adobe suite

Theoretical calculations

- Gaussian
- Siesta

Awards and Honors

1. State board scholarship for School toppers – J&K State Board of School Education (2010)
2. B.Sc. College Topper – Govt. Degree College Bijbehara, University of Kashmir (2015)
3. Merit Scholarship for Master's Program in Chemistry(1st rank in University entrance exam) – University of Kashmir (2015)
4. J&K State Level Eligibility Test for Assistant Professor (SLET) – 2018
5. Gold Medalist (M.Sc. Chemistry) – University of Kashmir (2018)
6. Graduate Aptitude Test in Engineering (GATE) Qualified – 2018
7. Council of Scientific and Industrial Research(CSIR)-JRF NET (Rank 63) – 2019
8. DST Inspire Fellowship for Research Program in Chemistry – 2018 (Provisional), 2019 (Final)
9. Prime Minister's Research Fellowship (PMRF) – Ministry of Education, Govt. of India - Most prestigious doctoral fellowship in country-August 2020
10. Vasudevamurthy-Soundararajan Award for Best Coursework Performance – IISc, 2019–20
11. Best Poster Presentation Award – International Conference on Spin in Molecular Systems, IISc (2022)
12. Vasudevamurthy-Soundararajan Award for Best Student Seminar – IISc, 2021–22
13. Best Oral Presentation Award – IPC-Day Symposium 2023, IISc Bangalore
14. Best Poster Presentation Award – International Conference on Recent Advances in Materials (RAM-90), JNCASR (2023)

Languages

English, Urdu, Kashmiri
